

Package ‘ClinicalCharacteristics’

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Description A tool to characterize cohorts using a table shell approach.

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dplyr,
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readr,
tibble,
tidyr,
here,
lubridate,
reactable,
methods

Additional_repositories <https://OHDSI.github.io/drat>

Suggests knitr,
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VignetteBuilder knitr

R topics documented:

addDefaultEthnicityLineItems	3
addDefaultGenderLineItems	3
adherentPresenceStat	4
age10yrGrp	4
age5yrGrp	4

ageCharBreaks	5
ageCharCts	5
anyCountBreaksStat	5
anyCountCtsStat	6
anyPresenceStat	6
anyScore	6
Breaks	7
BreaksStrategy	7
BuildOptions	8
cohortFollowupTime	10
CohortInfo	10
CohortLineItem	11
ConceptSetGroupLineItem	12
ConceptSetLineItem	13
ContinuousDistribution	14
createCohortInfo	15
createCohortLineItem	15
createCohortLineItemBatch	16
createConceptSetGroupLineItem	16
createConceptSetLineItem	17
createConceptSetLineItemBatch	18
createDemographicLineItem	19
createExecutionSettings	19
createSourceConceptSetLineItem	20
createSourceConceptSetLineItemBatch	21
createTableShell	21
defaultTableShellBuildOptions	22
defaultYearGrp	23
DemographicAge	24
DemographicCohortTime	25
DemographicConcept	25
DemographicIndexYear	26
DemographicLineItem	27
DemographicLocation	28
DemographicPayerType	29
DemographicRace	30
ExecutionSettings	31
femaleGender	32
generateTableShell	33
indexYear	33
IntervalRate	34
LineItem	34
lineItems	36
lookupSourceConcepts	36
maleGender	37
monthlyRate	37
newConceptBreaks	37
newValueBreaks	38
observedCountBreaksStat	38
observedCountCtsStat	39
observedPresenceStat	39
observedTimeToFirst	39

<i>addDefaultEthnicityLineItems</i>	3
-------------------------------------	---

parseCohortInfoFromDf	40
payerType	40
personLocation	41
Presence	41
quanCharlsonComorbidityScore	42
raceCategory	42
reviewTableShellSql	43
saveTableShellResults	43
Score	44
sourceConceptSet	45
SourceConceptSetLineItem	45
Statistic	46
TableShell	47
timeInterval	49
TimeIntervalClass	49
timeToFirst	50
yearlyRate	51

Index	52
--------------	-----------

addDefaultEthnicityLineItems

Convenience function to add default ethnicity line items

Description

Convenience function to add default ethnicity line items

Usage

addDefaultEthnicityLineItems()

Value

a list of line items for default ethnicity categories (hispanic, not hispanic, not reported)

addDefaultGenderLineItems

Convenience function to add male and female line items for demographic characterization

Description

Convenience function to add male and female line items for demographic characterization

Usage

addDefaultGenderLineItems()

Value

a list of two line items for male and female gender

adherentPresenceStat	<i>Adherent Presence Stat</i>
----------------------	-------------------------------

Description

Create a presence stat where only occurrence during the observation period are valid and the denominator are those who only adhere to the observation period

Usage

```
adherentPresenceStat()
```

Value

A presence stat object

age10yrGrp	<i>Create a breaks Strategy object for age into 10 year groups</i>
------------	--

Description

Create a breaks Strategy object for age into 10 year groups

Usage

```
age10yrGrp()
```

Value

A BreaksStrategy object with defaults assumptions for 10 year age groups

age5yrGrp	<i>Create a breaks Strategy object for age into 5 year groups</i>
-----------	---

Description

Create a breaks Strategy object for age into 5 year groups

Usage

```
age5yrGrp()
```

Value

A BreaksStrategy object with defaults assumptions for 5 year age groups

ageCharBreaks	Create a age statistic with breaks
---------------	------------------------------------

Description

Create a age statistic with breaks

Usage

```
ageCharBreaks(breaks)
```

Arguments

breaks	a breaksStrategy object dictating how to classify counts into categories
--------	--

Value

A DemographicAge Statistic class object with breaks

ageCharCts	Create a age statistic as continuous
------------	--------------------------------------

Description

Create a age statistic as continuous

Usage

```
ageCharCts()
```

Value

A DemographicAge Statistic class object as continuous

anyCountBreaksStat	Any Count with Breaks
--------------------	-----------------------

Description

Create a count stat with breaks where any occurrence is valid.

Usage

```
anyCountBreaksStat(breaks)
```

Arguments

breaks	a breaksStrategy object dictating how to classify counts into categories. If null then this defaults to a continuous distribution
--------	---

Value

A stat object breaks

anyCountCtsStat	<i>Any Count Continuous</i>
-----------------	-----------------------------

Description

Create a count stat where any occurrence is valid.

Usage

anyCountCtsStat()

Value

A stat object continuousDistribution

anyPresenceStat	<i>Any Presence Stat</i>
-----------------	--------------------------

Description

Create a presence stat where any occurrence is valid

Usage

anyPresenceStat()

Value

A presence stat object

anyScore	<i>Any Score</i>
----------	------------------

Description

Create score statistic

Usage

anyScore(weight)

Value

A stat object for a scoreTransformation

Breaks

Breaks Statistic

Description

A statistic that converts a continuous value to a categorical value by grouping the number of events into discrete buckets.

Super class

`ClinicalCharacteristics::Statistic` -> Breaks

Methods

Public methods:

- `Breaks$new()`
- `Breaks$clone()`

Method `new()`:

Usage:

`Breaks$new(personLine, breaks)`

Arguments:

`personLine` the means of converting occurrences to a single event per patient. For presence this could be any, observed or adherent
`breaks` a breaks strategy object to categorize results

Method `clone()`: The objects of this class are cloneable with this method.

Usage:

`Breaks$clone(deep = FALSE)`

Arguments:

`deep` Whether to make a deep clone.

BreaksStrategy

BreaksStrategy

Description

An R6 class to define a BreaksStrategy object

Active bindings

`name` the name of the breaks strategy

`type` the type of breaks strategy. Could be 'value' or 'concept'

`labels` A character vector used to label each break interval

`breaks` a vector with cut points

Methods

Public methods:

- `BreaksStrategy$new()`
- `BreaksStrategy$makeCaseWhenSql()`
- `BreaksStrategy$clone()`

Method `new()`:

Usage:

`BreaksStrategy$new(name, labels, breaks, type)`

Arguments:

`name` the name of the breaks strategy

`labels` a character vector indicating how to label each break interval

`breaks` a vector with cut points

`type` the type of breaks strategy. Could be 'value' or 'concept'

Method `makeCaseWhenSql()`: Generate SQL code for a CASE WHEN statement based on the break strategy

Usage:

`BreaksStrategy$makeCaseWhenSql(ordinalId)`

Arguments:

`ordinalId` the order identifier of the line item in the table shell

Method `clone()`: The objects of this class are cloneable with this method.

Usage:

`BreaksStrategy$clone(deep = FALSE)`

Arguments:

`deep` Whether to make a deep clone.

BuildOptions

BuildOptions

Description

An R6 class to define build options for the tableShell

Active bindings

`codesetTempTable` table name for codeset table

`sourceCodesetTempTable` table name for source codeset table

`timeWindowTempTable` table name for time windows

`targetCohortTempTable` table name for target cohorts

`tsMetaTempTable` table name for table shell meta

`conceptSetOccurrenceTempTable` table name for concept set occurrence table

`cohortOccurrenceTempTable` table name for cohort occurrence table

`patientLevelDataTempTable` table name for patient level data

patientLevelTableShellTempTable table name for patient level data table merged with ts meta
 categoricalSummaryTempTable table name for categorical summary table
 continuousSummaryTempTable table name for continuous summary table
 cohortAnalysisType toggle to choose if using cohort era or start date
 continuousEventType toggle to choose if you are summarizing all pop or just events

Methods

Public methods:

- [BuildOptions\\$new\(\)](#)
- [BuildOptions\\$clone\(\)](#)

Method new():

Usage:

```
BuildOptions$new(
  codesetTempTable = NULL,
  sourceCodesetTempTable = NULL,
  timeWindowTempTable = NULL,
  targetCohortTempTable = NULL,
  tsMetaTempTable = NULL,
  conceptSetOccurrenceTempTable = NULL,
  cohortOccurrenceTempTable = NULL,
  patientLevelDataTempTable = NULL,
  patientLevelTableShellTempTable = NULL,
  categoricalSummaryTempTable = NULL,
  continuousSummaryTempTable = NULL,
  cohortAnalysisType = NULL,
  continuousEventType = NULL
)
```

Arguments:

codesetTempTable the name of the codeset table used in execution. Defaults as a temp table #codeset

sourceCodesetTempTable the name of the source codeset table used in execution

timeWindowTempTable the name of the time Window table used in execution. Defaults as a temp table #time_windows

targetCohortTempTable the name of the target cohort table used in execution. Defaults as a temp table #target_cohorts

tsMetaTempTable the name of the table shell meta table used in execution. Defaults as a temp table #ts_meta

conceptSetOccurrenceTempTable the name of the concept set occurrence table used in execution. Defaults as a temp table #concept_set_occ

cohortOccurrenceTempTable the name of the cohort occurrence table used in execution. Defaults as a temp table #cohort_occ

patientLevelDataTempTable the name of the patient level data table used in execution. Note this does not contain info of the table shell. Defaults as a temp table #patient_data

patientLevelTableShellTempTable the name of the patient level data table with additional meta info used in execution. Defaults as a temp table #pat_ts_tab

categoricalSummaryTempTable the name of the categorical summary table used in execution. Defaults as a temp table #categorical_table

`continuousSummaryTempTable` the name of the continuous summary table used in execution.
 Defaults as a temp table `#continuous_table`
`cohortAnalysisType` a toggle specifying if in a cohort Char whether to use the cohort era
 ('era') or just the start date ('startDate')
`continuousEventType` a toggle specifying if the continuous aggregation should be on the full
 population or persons with events

Method `clone()`: The objects of this class are cloneable with this method.

Usage:

`BuildOptions$clone(deep = FALSE)`

Arguments:

`deep` Whether to make a deep clone.

<code>cohortFollowupTime</code>	<i>Create a cohort follow up time characteristic</i>
---------------------------------	--

Description

Create a cohort follow up time characteristic

Usage

`cohortFollowupTime()`

Value

A `DemographicCohortTime` Statistic class object

<code>CohortInfo</code>	<i>CohortInfo</i>
-------------------------	-------------------

Description

An R6 class to define a Cohort Info object. `CohortInfo` objects do not maintain any execution settings, just the id and name

Methods

Public methods:

- `CohortInfo$new()`
- `CohortInfo$getId()`
- `CohortInfo$getName()`
- `CohortInfo$cohortDetails()`
- `CohortInfo$clone()`

Method `new()`:

Usage:

```
CohortInfo$new(id, name)
```

Arguments:

id the cohort definition id

name the name of the cohort definition

Method getId(): get the cohort id

Usage:

```
CohortInfo$getId()
```

Method getName(): get the cohort name

Usage:

```
CohortInfo$getName()
```

Method cohortDetails(): print the cohort details

Usage:

```
CohortInfo$cohortDetails()
```

Method clone(): The objects of this class are cloneable with this method.

Usage:

```
CohortInfo$clone(deep = FALSE)
```

Arguments:

deep Whether to make a deep clone.

CohortLineItem

CohortLineItem

Description

An R6 class to define a CohortLineItem

Super class

`ClinicalCharacteristics::LineItem` -> CohortLineItem

Methods

Public methods:

- `CohortLineItem$new()`
- `CohortLineItem$clone()`

Method new():

Usage:

```
CohortLineItem$new(
  sectionLabel,
  domainTable,
  covariateCohort,
  timeInterval,
  statistic
)
```

Arguments:

sectionLabel a label for the table shell section

domainTable the domain table in the cdm

covariateCohort a CohortInfo class with cohorts for covariates

timeInterval a time interval class object to determine the time frame to consider the analytic

statistic a Statistic Class object used to determine what type of analytic should be done for the line item

Method clone(): The objects of this class are cloneable with this method.

Usage:

```
CohortLineItem$clone(deep = FALSE)
```

Arguments:

deep Whether to make a deep clone.

ConceptSetGroupLineItem

ConceptSetGroupLineItem

Description

An R6 class to define a ConceptSetGroupLineItem

Super class

[ClinicalCharacteristics::LineItem](#) -> ConceptSetGroupLineItem

Methods**Public methods:**

- [ConceptSetGroupLineItem\\$new\(\)](#)
- [ConceptSetGroupLineItem\\$grabConceptSet\(\)](#)
- [ConceptSetGroupLineItem\\$clone\(\)](#)

Method new():*Usage:*

```
ConceptSetGroupLineItem$new(
  sectionLabel,
  groupLabel,
  conceptSets,
  domainTables,
  timeInterval,
  statistic
)
```

Arguments:

sectionLabel a label for the table shell section

groupLabel a label for the group

conceptSets a group of concept sets

domainTables the domain tables in the cdm
 timeInterval a time interval class object to determine the time frame to consider the analytic
 statistic a Statistic Class object used to determine what type of analytic should be done for
 the line item

Method grabConceptSet(): retrieve the concept sets

Usage:

```
ConceptSetGroupLineItem$grabConceptSet()
```

Method clone(): The objects of this class are cloneable with this method.

Usage:

```
ConceptSetGroupLineItem$clone(deep = FALSE)
```

Arguments:

deep Whether to make a deep clone.

ConceptSetLineItem	<i>ConceptSetLineItem</i>
--------------------	---------------------------

Description

An R6 class to define a ConceptSetLineItem

Super class

[ClinicalCharacteristics::LineItem](#) -> ConceptSetLineItem

Methods

Public methods:

- [ConceptSetLineItem\\$new\(\)](#)
- [ConceptSetLineItem\\$grabConceptSet\(\)](#)
- [ConceptSetLineItem\\$clone\(\)](#)

Method new():

Usage:

```
ConceptSetLineItem$new(  
  sectionLabel,  
  domainTable,  
  conceptSet,  
  timeInterval,  
  statistic  
)
```

Arguments:

sectionLabel a label for the table shell section

domainTable the domain table in the cdm

conceptSet a concept set class from Capr

timeInterval a time interval class object to determine the time frame to consider the analytic

statistic a Statistic Class object used to determine what type of analytic should be done for the line item grabConceptSet

Method grabConceptSet(): helper to pull concept Capr class items

Usage:

ConceptSetLineItem\$grabConceptSet()

Method clone(): The objects of this class are cloneable with this method.

Usage:

ConceptSetLineItem\$clone(deep = FALSE)

Arguments:

deep Whether to make a deep clone.

ContinuousDistribution

Continuous Distribution Statistic

Description

A statistic that summarizes the number of occurrences as continuous value using mean, standard deviation and order statistics

Super class

`ClinicalCharacteristics::Statistic -> ContinuousDistribution`

Methods

Public methods:

- `ContinuousDistribution$new()`
- `ContinuousDistribution$clone()`

Method new():

Usage:

ContinuousDistribution\$new(personLine)

Arguments:

personLine the means of converting occurrences to a single event per patient. For presence this could be any, observed or adherent

Method clone(): The objects of this class are cloneable with this method.

Usage:

ContinuousDistribution\$clone(deep = FALSE)

Arguments:

deep Whether to make a deep clone.

createCohortInfo	Create a CohortInfo object for a cohort and set its attributes
------------------	--

Description

Create a CohortInfo object for a cohort and set its attributes

Usage

```
createCohortInfo(id, name)
```

Arguments

id	The ID of the cohort
name	The name of the cohort

Value

A CohortInfo object

createCohortLineItem	Create a cohort line item and set its attributes
----------------------	--

Description

Create a cohort line item and set its attributes

Usage

```
createCohortLineItem(  
  sectionLabel = NA_character_,  
  covariateCohort,  
  cohortTable,  
  timeInterval,  
  statistic  
)
```

Arguments

sectionLabel	(OPTIONAL) The name of the line item (if not provided, the name will be set to the cohort name from the CohortInfo object)
timeInterval	The TimeIntervalClass object used for the line item
statistic	The Statistic object to be used to evaluate the line item
cohort	A CohortInfo object

Value

A CohortLineItem object

```
createCohortLineItemBatch
```

Create a batch of cohort line items from a list of CohortInfo objects.

Description

The name of each line item will be set to the name of its cohort from the CohortInfo object.

Usage

```
createCohortLineItemBatch(
  sectionLabel,
  covariateCohorts,
  cohortTable,
  statistic,
  timeIntervals
)
```

Arguments

sectionLabel	The name of the cohort batch
statistic	The Statistic object to be used to evaluate the line items
timeIntervals	A list of TimeIntervalClass objects
cohorts	A list of CohortInfo objects

Value

A list of CohortLineItem objects

```
createConceptSetGroupLineItem
```

Create a concept set group item and set its attributes

Description

Create a concept set group item and set its attributes

Usage

```
createConceptSetGroupLineItem(
  sectionLabel = NA_character_,
  groupLabel,
  conceptSets,
  domainTables,
  timeInterval,
  statistic
)
```


Arguments

sectionLabel	(OPTIONAL) The name of the line item (if not provided, the name will be the same as the group label)
groupLabel	the label of the group
conceptSets	A list of Capr concept set object
domainTables	a vector of domains corresponding to the concept set
timeInterval	The TimeIntervalClass object used for the line item
statistic	The Statistic object to be used to evaluate the line item

Value

A CohortLineItem object

```
createConceptSetLineItem
```

Create a concept set line item and set its attributes

Description

Create a concept set line item and set its attributes

Usage

```
createConceptSetLineItem(
  sectionLabel = NA_character_,
  domain,
  conceptSet,
  timeInterval,
  statistic
)
```

Arguments

sectionLabel	(OPTIONAL) The name of the line item (if not provided, the name will be set to the Capr concept set name)
domain	The domain of the concept set (must be one of 'Condition', 'Drug', 'Procedure', 'Observation', 'Measurement', 'Device')
conceptSet	The Capr concept set object
timeInterval	The Time Interval object used for the line item
statistic	The Statistic object to be used to evaluate the line item
sourceConceptSet	(OPTIONAL) A Capr concept set of source concept IDs to use to limit the concept set
typeConceptIds	(OPTIONAL) A list of type concept IDs to use to limit the concept set
visitOccurrenceConceptIds	(OPTIONAL) A list of visit occurrence concept IDs to use to limit the concept set

Value

A ConceptSetLineItem object

createConceptSetLineItemBatch

Create a batch of concept set line items from a list of Capr concept sets.

Description

The name of each line item will be set to the name of its Capr concept set. All line items will use the same statistic, domain, type concepts, and visit concepts. It is not possible to specify source concept IDs.

Usage

```
createConceptSetLineItemBatch(
    sectionLabel,
    domain,
    conceptSets,
    timeIntervals,
    statistic
)
```

Arguments

sectionLabel	The name of the concept set batch
domain	The domain of the concept sets (must be one of 'Condition', 'Drug', 'Procedure', 'Observation', 'Measurement', 'Device')
conceptSets	A list of concept set Capr objects
timeIntervals	A list of TimeIntervalClass objects
statistic	The Statistic object to be used to evaluate the line items
typeConceptIds	(OPTIONAL) A list of type concept IDs to use to limit the concept set
visitOccurrenceConceptIds	(OPTIONAL) A list of visit occurrence concept IDs to use to limit the concept set

Value

A list of ConceptSetLineItem objects

```
createDemographicLineItem
```

Create a demographic line item and set its attributes

Description

Create a demographic line item and set its attributes

Usage

```
createDemographicLineItem(statistic)
```

Arguments

statistic	The Statistic object to be used to evaluate the line item
-----------	---

Value

A DemographicLineItem object

```
createExecutionSettings
```

Create an ExecutionSettings object and set its attributes

Description

Create an ExecutionSettings object and set its attributes

Usage

```
createExecutionSettings(
  connectionDetails,
  connection = NULL,
  cdmDatabaseSchema,
  workDatabaseSchema,
  tempEmulationSchema,
  cohortTable,
  cdmSourceName
)
```

Arguments

connectionDetails	A DatabaseConnector connectionDetails object (optional if connection is specified)
connection	A DatabaseConnector connection object (optional if connectionDetails is specified)
cdmDatabaseSchema	The schema of the OMOP CDM database

workDatabaseSchema	The schema to which results will be written
tempEmulationSchema	Some database platforms like Oracle and Snowflake do not truly support temp tables. To emulate temp tables, provide a schema with write privileges where temp tables can be created.
cohortTable	The name of the table where the cohort(s) are stored
cdmSourceName	A human-readable name for the OMOP CDM source

Value

An ExecutionSettings object

createSourceConceptSetLineItem
Create a source concept set line item and set its attributes

Description

Create a source concept set line item and set its attributes

Usage

```
createSourceConceptSetLineItem(
  sectionLabel = NA_character_,
  domain,
  sourceConceptSet,
  timeInterval,
  statistic,
  typeConceptIds = c()
)
```

Arguments

sectionLabel	(OPTIONAL) The name of the line item (if not provided, the name will be set to the Capr concept set name)
domain	The domain of the concept set (must be one of 'Condition', 'Drug', 'Procedure', 'Observation', 'Measurement', 'Device')
sourceConceptSet	A SourceConcept R6 object created using the sourceConceptSet function
timeInterval	The Time Interval object used for the line item
statistic	The Statistic object to be used to evaluate the line item
typeConceptIds	(OPTIONAL) A list of type concept IDs to use to limit the concept set

Value

A SourceConceptSetLineItem object

createSourceConceptSetLineItemBatch	<i>Create a batch of source concept set line items from a list of SourceConceptSet classes.</i>
-------------------------------------	---

Description

Create a batch of source concept set line items from a list of SourceConceptSet classes.

Usage

```
createSourceConceptSetLineItemBatch(  
  sectionLabel,  
  domain,  
  sourceConceptSets,  
  timeIntervals,  
  statistic,  
  typeConceptIds = c()  
)
```

Arguments

- sectionLabel (OPTIONAL) The name of the line item (if not provided, the name will be set to the Capr concept set name)
- domain The domain of the concept set (must be one of 'Condition', 'Drug', 'Procedure', 'Observation', 'Measurement', 'Device')
- timeIntervals A list of TimeIntervalClass objects
- statistic The Statistic object to be used to evaluate the line item
- typeConceptIds (OPTIONAL) A list of type concept IDs to use to limit the concept set
- sourceConceptSet A list of SourceConcept R6 object created using the sourceConceptSet function

Value

A list of SourceConceptSetLineItem objects

createTableShell	<i>Create Table Shell</i>
------------------	---------------------------

Description

Create an empty TableShell object and set its title

Usage

```
createTableShell(title, targetCohorts, lineItems)
```

Arguments

title	The title of the TableShell
targetCohorts	A list of TargetCohort objects
lineItems	A list of lineItem objects

Value

A TableShell object

defaultTableShellBuildOptions

Default build options to generate table shell

Description

Default build options to generate table shell

Usage

```
defaultTableShellBuildOptions(
  codesetTempTable = "#codeset",
  sourceCodesetTempTable = "#source_codeset",
  timeWindowTempTable = "#time_windows",
  targetCohortTempTable = "#target_cohorts",
  tsMetaTempTable = "#ts_meta",
  conceptSetOccurrenceTempTable = "#concept_set_occ",
  cohortOccurrenceTempTable = "#cohort_occ",
  patientLevelDataTempTable = "#patient_data",
  patientLevelTableShellTempTable = "#pat_ts_tab",
  categoricalSummaryTempTable = "#categorical_table",
  continuousSummaryTempTable = "#continuous_table",
  cohortAnalysisType = c("era", "startDate"),
  continuousEventType = c("all", "events")
)
```

Arguments

codesetTempTable	the name of the codeset table used in execution. Defaults as a temp table #code-set
timeWindowTempTable	the name of the time Window table used in execution. Defaults as a temp table #time_windows
targetCohortTempTable	the name of the target cohort table used in execution. Defaults as a temp table #target_cohorts
tsMetaTempTable	the name of the table shell meta table used in execution. Defaults as a temp table #ts_meta

conceptSetOccurrenceTempTable	the name of the concept set occurrence table used in execution. Defaults as a temp table #concept_set_occ
cohortOccurrenceTempTable	the name of the cohort occurrence table used in execution. Defaults as a temp table #cohort_occ
patientLevelDataTempTable	the name of the patient level data table used in execution. Note this does not contain info of the table shell. Defaults as a temp table #patient_data
patientLevelTableShellTempTable	the name of the patient level data table with additional meta info used in execution. Defaults as a temp table #pat_ts_tab
categoricalSummaryTempTable	the name of the categorical summary table used in execution. Defaults as a temp table #categorical_table
continuousSummaryTempTable	the name of the continuous summary table used in execution. Defaults as a temp table #continuous_table
cohortAnalysisType	a toggle specifying if in a cohort Char whether to use the cohort era ('era') or just the start date ('startDate')
continuousEventType	a toggle specifying if the continuous aggregation should be on the full population or persons with events

Value

A BuildOptions object

defaultYearGrp	<i>Create a breaks Strategy object for year</i>
----------------	---

Description

Create a breaks Strategy object for year

Usage

```
defaultYearGrp(startYear = NULL)
```

Arguments

startYear the year to start the year group sequence. By default this is the year 2000

Value

A BreaksStrategy object with defaults assumptions for 5 year age groups

DemographicAge

*Demographic Age Statistic***Description**

A Demographic Statistic that calculates age from the person table

Super class

`ClinicalCharacteristics::Statistic` -> DemographicAge

Methods**Public methods:**

- `DemographicAge$new()`
- `DemographicAge$getDemoLabel()`
- `DemographicAge$modifyBreaksLabels()`
- `DemographicAge$clone()`

Method `new()`:

Usage:

`DemographicAge$new(statType, aggType, demoCategory, breaks = NULL)`

Arguments:

`statType` the type of statistic

`aggType` the way the metric is reported either categorical or continuous

`demoCategory` the name of the demographic category

`breaks` a breaks strategy object to categorize results

Method `getDemoLabel()`: retrieve the demographic label

Usage:

`DemographicAge$getDemoLabel()`

Method `modifyBreaksLabels()`: update the breaks labels within the statistics class

Usage:

`DemographicAge$modifyBreaksLabels(newLabels)`

Arguments:

`newLabels` a character string of new labels for the breaks

Method `clone()`: The objects of this class are cloneable with this method.

Usage:

`DemographicAge$clone(deep = FALSE)`

Arguments:

`deep` Whether to make a deep clone.

DemographicCohortTime *Demographic Cohort Time Statistic*

Description

A Demographic Statistic that calculates the time (in days) in the target cohort

Super class

`ClinicalCharacteristics::Statistic` -> DemographicCohortTime

Methods

Public methods:

- `DemographicCohortTime$new()`
- `DemographicCohortTime$getDemoLabel()`
- `DemographicCohortTime$clone()`

Method `new()`: initialize cohort time stat

Usage:

`DemographicCohortTime$new()`

Method `getDemoLabel()`: retrieve the demographic label

Usage:

`DemographicCohortTime$getDemoLabel()`

Method `clone()`: The objects of this class are cloneable with this method.

Usage:

`DemographicCohortTime$clone(deep = FALSE)`

Arguments:

`deep` Whether to make a deep clone.

DemographicConcept *Demographic Concept Statistic*

Description

A Demographic Statistic that considers concepts in person table

Super class

`ClinicalCharacteristics::Statistic` -> DemographicConcept

Methods**Public methods:**

- [DemographicConcept\\$new\(\)](#)
- [DemographicConcept\\$getConceptColumn\(\)](#)
- [DemographicConcept\\$getDemoLabel\(\)](#)
- [DemographicConcept\\$getConceptId\(\)](#)
- [DemographicConcept\\$clone\(\)](#)

Method new():*Usage:*

DemographicConcept\$new(demoCategory, demoLine, conceptColumn, conceptId)

Arguments:

demoCategory the category name of the demographic

demoLine the line item name of the demographic concept

conceptColumn the name of column in the person table to extract demographic concept

conceptId the concept to search for in the concept column

Method getConceptColumn(): retrieve the concept column*Usage:*

DemographicConcept\$getConceptColumn()

Method getDemoLabel(): create a label for the demographic concept*Usage:*

DemographicConcept\$getDemoLabel()

Method getConceptId(): retrieve the concept id*Usage:*

DemographicConcept\$getConceptId()

Method clone(): The objects of this class are cloneable with this method.*Usage:*

DemographicConcept\$clone(deep = FALSE)

Arguments:

deep Whether to make a deep clone.

DemographicIndexYear *Demographic Index Year Statistic*

Description

A Demographic Statistic that retrieves the index year for each patient

Super class[ClinicalCharacteristics::Statistic](#) -> DemographicIndexYear

Methods**Public methods:**

- [DemographicIndexYear\\$new\(\)](#)
- [DemographicIndexYear\\$getDemoLabel\(\)](#)
- [DemographicIndexYear\\$modifyBreaksLabels\(\)](#)
- [DemographicIndexYear\\$clone\(\)](#)

Method `new()`:*Usage:*`DemographicIndexYear$new(breaks)`*Arguments:*`breaks` a breaks strategy object to categorize results**Method** `getDemoLabel()`: retrieve the demographic label*Usage:*`DemographicIndexYear$getDemoLabel()`**Method** `modifyBreaksLabels()`: update the breaks labels within the statistics class*Usage:*`DemographicIndexYear$modifyBreaksLabels(newLabels)`*Arguments:*`newLabels` a character string of new labels for the breaks**Method** `clone()`: The objects of this class are cloneable with this method.*Usage:*`DemographicIndexYear$clone(deep = FALSE)`*Arguments:*`deep` Whether to make a deep clone.

DemographicLineItem	<i>DemographicLineItem</i>
---------------------	----------------------------

Description

An R6 class to handle a Demographic line item

Super class[ClinicalCharacteristics::LineItem](#) -> DemographicLineItem

Methods**Public methods:**

- [DemographicLineItem\\$new\(\)](#)
- [DemographicLineItem\\$clone\(\)](#)

Method new():*Usage:*

DemographicLineItem\$new(Statistic = Statistic)

Arguments:

Statistic a Statistic Class object used to determine what type of analytic should be done for the line item

Method clone(): The objects of this class are cloneable with this method.

Usage:

DemographicLineItem\$clone(deep = FALSE)

Arguments:

deep Whether to make a deep clone.

DemographicLocation *Demographic Location Statistic*

Description

A Demographic Statistic that retrieves and categorizes the location of the persons in the target cohort

Super class

[ClinicalCharacteristics::Statistic](#) -> DemographicLocation

Methods**Public methods:**

- [DemographicLocation\\$new\(\)](#)
- [DemographicLocation\\$getDemoLabel\(\)](#)
- [DemographicLocation\\$modifyBreaksLabels\(\)](#)
- [DemographicLocation\\$clone\(\)](#)

Method new():*Usage:*

DemographicLocation\$new(breaks)

Arguments:

breaks a breaks strategy object to categorize results

Method getDemoLabel(): retrieve the demographic label

Usage:

DemographicLocation\$getDemoLabel()

Method modifyBreaksLabels(): update the breaks labels within the statistics class

Usage:

DemographicLocation\$modifyBreaksLabels(newLabels)

Arguments:

newLabels a character string of new labels for the breaks

Method clone(): The objects of this class are cloneable with this method.

Usage:

DemographicLocation\$clone(deep = FALSE)

Arguments:

deep Whether to make a deep clone.

DemographicPayerType *Demographic Payer Statistic*

Description

A Demographic Statistic that retrieves and categorizes the payer type from the payer plan period table

Super class

[ClinicalCharacteristics::Statistic](#) -> DemographicPayerType

Methods

Public methods:

- [DemographicPayerType\\$new\(\)](#)
- [DemographicPayerType\\$getDemoLabel\(\)](#)
- [DemographicPayerType\\$modifyBreaksLabels\(\)](#)
- [DemographicPayerType\\$clone\(\)](#)

Method new():

Usage:

DemographicPayerType\$new(breaks)

Arguments:

breaks a breaks strategy object to categorize results

Method getDemoLabel(): retrieve the demographic label

Usage:

DemographicPayerType\$getDemoLabel()

Method modifyBreaksLabels(): update the breaks labels within the statistics class

Usage:

DemographicPayerType\$modifyBreaksLabels(newLabels)

Arguments:

`newLabels` a character string of new labels for the breaks

Method `clone()`: The objects of this class are cloneable with this method.

Usage:

`DemographicPayerType$clone(deep = FALSE)`

Arguments:

`deep` Whether to make a deep clone.

DemographicRace

Demographic Race Statistic

Description

A Demographic Statistic that retrieves and categorizes the patient race from the person table

Super class

`ClinicalCharacteristics::Statistic` -> DemographicRace

Methods

Public methods:

- `DemographicRace$new()`
- `DemographicRace$getDemoLabel()`
- `DemographicRace$modifyBreaksLabels()`
- `DemographicRace$clone()`

Method `new()`:

Usage:

`DemographicRace$new(breaks)`

Arguments:

`breaks` a breaks strategy object to categorize results

Method `getDemoLabel()`: retrieve the demographic label

Usage:

`DemographicRace$getDemoLabel()`

Method `modifyBreaksLabels()`: update the breaks labels within the statistics class

Usage:

`DemographicRace$modifyBreaksLabels(newLabels)`

Arguments:

`newLabels` a character string of new labels for the breaks

Method `clone()`: The objects of this class are cloneable with this method.

Usage:

`DemographicRace$clone(deep = FALSE)`

Arguments:

`deep` Whether to make a deep clone.

ExecutionSettings	<i>ExecutionSettings</i>
-------------------	--------------------------

Description

An R6 class to define an ExecutionSettings object

Active bindings

cdmDatabaseSchema the schema containing the OMOP CDM

workDatabaseSchema the schema containing the cohort table

tempEmulationSchema the schema needed for temp tables

cohortTable the table containing the cohorts

cdmSourceName the name of the source data of the cdm

Methods

Public methods:

- [ExecutionSettings\\$new\(\)](#)
- [ExecutionSettings\\$getDbms\(\)](#)
- [ExecutionSettings\\$connect\(\)](#)
- [ExecutionSettings\\$disconnect\(\)](#)
- [ExecutionSettings\\$getConnection\(\)](#)
- [ExecutionSettings\\$clone\(\)](#)

Method new():

Usage:

```
ExecutionSettings$new(
  connectionDetails = NULL,
  connection = NULL,
  cdmDatabaseSchema = NULL,
  workDatabaseSchema = NULL,
  tempEmulationSchema = NULL,
  cohortTable = NULL,
  cdmSourceName = NULL
)
```

Arguments:

connectionDetails a connectionDetails object

connection a connection to a dbms

cdmDatabaseSchema The schema of the OMOP CDM database

workDatabaseSchema The schema to which results will be written

tempEmulationSchema Some database platforms like Oracle and Snowflake do not truly support temp tables. To emulate temp tables, provide a schema with write privileges where temp tables can be created.

cohortTable The name of the table where the cohort(s) are stored

cdmSourceName A human-readable name for the OMOP CDM source

Method getDbms(): extract the dbms dialect

Usage:

ExecutionSettings\$dbms()

Method connect(): connect to dbms

Usage:

ExecutionSettings\$connect()

Method disconnect(): disconnect from dbms

Usage:

ExecutionSettings\$disconnect()

Method getConnection(): retrieve the connection object

Usage:

ExecutionSettings\$getConnection()

Method clone(): The objects of this class are cloneable with this method.

Usage:

ExecutionSettings\$clone(deep = FALSE)

Arguments:

deep Whether to make a deep clone.

femaleGender

Create a female concept stat

Description

Create a female concept stat

Usage

femaleGender()

Value

A DemographicConcept Statistic class object indicating a female concept

generateTableShell	<i>Function to generate results for the table shell object</i>
--------------------	--

Description

Function to generate results for the table shell object

Usage

```
generateTableShell(tableShell, executionSettings, buildOptions = NULL)
```

Arguments

tableShell	The TableShell object to used for generation
executionSettings	The ExecutionSettings object used to generate table shell
buildOptions	The BuildOptions object used to generate table shell

Value

A list containing a tibble for categorical and continuous results

indexYear	<i>Create an index year characteristic</i>
-----------	--

Description

Create an index year characteristic

Usage

```
indexYear(breaks = NULL)
```

Arguments

breaks	a breaksStrategy object dictating how to classify years into categories. By default this will do each year from 2000 to current day.
--------	--

Value

A DemographicIndexYear Statistic class object

IntervalRate	<i>Interval Rate Statistic</i>
--------------	--------------------------------

Description

A statistic that calculates the rate of occurrence by taking the number of events per person in the desired interval and dividing by the observed time during the interval. An interval rate can either be monthly or yearly.

Super class

`ClinicalCharacteristics::Statistic` -> IntervalRate

Methods

Public methods:

- `IntervalRate$new()`
- `IntervalRate$clone()`

Method new():

Usage:

`IntervalRate$new(interval)`

Arguments:

interval the type of interval to use for the rate. can be either monthly or yearly.

Method clone(): The objects of this class are cloneable with this method.

Usage:

`IntervalRate$clone(deep = FALSE)`

Arguments:

deep Whether to make a deep clone.

LineItem	<i>LineItem</i>
----------	-----------------

Description

An R6 class to define a LineItem object. A LineItem is a single, explicitly defined characterization to appear in a Section. Derived classes exist off of LineItems

Active bindings

ordinalId the order identifier of the line item in the table shell

sectionLabel a label for the table shell section

lineItemLabel a label for the line item

valueId the id for the line item; either a codeset id, a concept id or a -999 to indicate no true id

valueDescription the describer for the value id

domainTable the domain table in the cdm

lineItemClass the type of line item (ie Demographic, ConceptSet, SourceConceptSet, ConceptSetGroup, Cohort)

Methods**Public methods:**

- `LineItem$new()`
- `LineItem$getLineItemMeta()`
- `LineItem$getStatistic()`
- `LineItem$clone()`

Method new():*Usage:*

```
LineItem$new(
  sectionLabel,
  lineItemLabel = NA_character_,
  domainTable,
  lineItemClass,
  valueId = NA_integer_,
  valueDescription = NA_integer_,
  statistic,
  timeInterval = NULL
)
```

Arguments:

sectionLabel a label for the table shell section

lineItemLabel a label for the line item

domainTable the domain table in the cdm

lineItemClass the type of line item (ie Demographic, ConceptSet, SourceConceptSet, ConceptSetGroup, Cohort)

valueId the id for the line item; either a codeset id, a concept id or a -999 to indicate no true id

valueDescription the describer for the value id

statistic a Statistic Class object used to determine what type of analytic should be done for the line item

timeInterval a time interval class object to determine the time frame to consider the analytic

Method getLineItemMeta(): retrieve the line item meta information*Usage:*

```
LineItem$getLineItemMeta()
```

Method getStatistic(): retrieve the statistic class object*Usage:*

```
LineItem$getStatistic()
```

Method clone(): The objects of this class are cloneable with this method.*Usage:*

```
LineItem$clone(deep = FALSE)
```

Arguments:

deep Whether to make a deep clone.

lineItems	<i>Combine all lineItems to enter into the tableShell slot</i>
-----------	--

Description

Combine all lineItems to enter into the tableShell slot

Usage

```
lineItems(...)
```

Arguments

...	A list of lineItems created from various calls
-----	--

Value

a flattened list of lineItems

lookupSourceConcepts	<i>Function to look up source concepts in the OMOP Vocabulary</i>
----------------------	---

Description

Function to look up source concepts in the OMOP Vocabulary

Usage

```
lookupSourceConcepts(codes, vocabulary, executionSettings)
```

Arguments

codes	a character string of codes to search
vocabulary	the vocabulary to use in search of codes
executionSettings	The ExecutionSettings object used to connect to the dbms

Value

a tibble of four columns: conceptId, conceptName, conceptCode, vocabularyId

maleGender	<i>Create a male concept stat</i>
------------	-----------------------------------

Description

Create a male concept stat

Usage

```
maleGender()
```

Value

A DemographicConcept Statistic class object indicating a male concept

monthlyRate	<i>Create a monthly interval rate statistic</i>
-------------	---

Description

This statistic sums the number of occurrences of an event in a timeInterval and divides it by the time (modified by month) to construct a rate per patient. This can then be summarized as a continuous variable

Usage

```
monthlyRate()
```

Value

A stat object of class intervalRate

newConceptBreaks	<i>Create a breaks Strategy object for categorizing concepts</i>
------------------	--

Description

Create a breaks Strategy object for categorizing concepts

Usage

```
newConceptBreaks(name, breaks, labels)
```

Arguments

name	the name of the breaks
breaks	a vector with cut points to user
labels	a character vector indicating how to label the cut-point. Can stay NULL where a default label is given

Value

A BreaksStrategy object of type concept

newValueBreaks	<i>Create a breaks Strategy object for categorizing value</i>
----------------	---

Description

Create a breaks Strategy object for categorizing value

Usage

```
newValueBreaks(name, breaks, labels = NULL)
```

Arguments

name	the name of the breaks
breaks	a vector with cut points to user
labels	a character vector indicating how to label the cut-point. Can stay NULL where a default label is given

Value

A BreaksStrategy object of type value

observedCountBreaksStat	<i>Observed Count with Breaks</i>
-------------------------	-----------------------------------

Description

Create a count stat with breaks where only occurrence during the observation period are valid

Usage

```
observedCountBreaksStat(breaks)
```

Arguments

breaks	a breaksStrategy object dictating how to classify counts into categories. If null then this defaults to a continuous distribution
--------	---

Value

A stat object breaks

observedCountCtsStat	<i>Observed Count Continuous</i>
----------------------	----------------------------------

Description

Create a count stat where only occurrence during the observation period are valid

Usage

```
observedCountCtsStat()
```

Value

A stat object continuousDistribution

observedPresenceStat	<i>Observed Presence Stat</i>
----------------------	-------------------------------

Description

Create a presence stat where only occurrence during the observation period are valid

Usage

```
observedPresenceStat()
```

Value

A presence stat object

observedTimeToFirst	<i>Observed Time To First</i>
---------------------	-------------------------------

Description

Create a time to stat where only occurrence during the observation period is valid

Usage

```
observedTimeToFirst()
```

Value

A stat object continuousDistribution

parseCohortInfoFromDf	<i>Parse cohort info from a data frame</i>
-----------------------	--

Description

Parse cohort info from a data frame

Usage

```
parseCohortInfoFromDf(df)
```

Arguments

df	The data frame containing the information for the cohorts (id and name)
----	---

Value

A list of CohortInfo objects

payerType	<i>Create a payer type characteristic</i>
-----------	---

Description

Create a payer type characteristic

Usage

```
payerType(breaks = NULL)
```

Arguments

breaks	a breaksStrategy object dictating how to classify payer types into categories. by default this will use the Source of Payment Typology(SOPT) vocabulary
--------	---

Value

A DemographicPayerType Statistic class object

personLocation	<i>Create a location characteristic</i>
----------------	---

Description

Create a location characteristic

Usage

personLocation(breaks)

Arguments

breaks a breaksStrategy object dictating how to classify locations into categories.

Value

A DemographicLocation Statistic class object

Presence	<i>Presence Statistic</i>
----------	---------------------------

Description

A statistic that determines whether at least one clinical event was present during the specified time interval. It is summarized as a categorical value.

Super class

`ClinicalCharacteristics::Statistic` -> Presence

Methods**Public methods:**

- `Presence$new()`
- `Presence$clone()`

Method new():

Usage:

`Presence$new(personLine)`

Arguments:

personLine the means of converting occurrences to a single event per patient. For presence this could be any, observed or adherent

Method clone(): The objects of this class are cloneable with this method.

Usage:

`Presence$clone(deep = FALSE)`

Arguments:

deep Whether to make a deep clone.

```
quanCharlsonComorbidityScore
```

Convenience function to add quan charlson comorbidity score

Description

The Quan Charlson Comorbidity score is a measure for predicting 10 year survival. It is a modification to the Charlson Score by Quan et al (doi: 10.1097/01.mlr.0000182534.19832.83). The method presented in this packages follows the SNOMED adaption of Quan Charlson tested on OMOP CDM by Fortin et al (doi: 10.1186/s12911-022-02006-1). This function will add the elements needed for each comorbidity line item and the appropriate weights needed to convert the categorization of comorbidities into a score.

Usage

```
quanCharlsonComorbidityScore(timeWindow = NULL)
```

Arguments

`timeWindow` the interval to assess the comorbidity score, by default baseline it -365 to -1 days

Value

a list of line items for running quan charlson comorbidity score. This will determine the proportion of persons with each comorbidity and the overall score per patient in the cohort

```
raceCategory
```

Create a race characteristic

Description

Create a race characteristic

Usage

```
raceCategory(breaks = NULL)
```

Arguments

`breaks` a breaksStrategy object dictating how to classify race into categories. by default this will use custom race categories

Value

A DemographicRace Statistic class object

reviewTableShellSql	<i>Function that previews sql script used to generate results for table shell</i>
---------------------	---

Description

Function that previews sql script used to generate results for table shell

Usage

```
reviewTableShellSql(
  tableShell,
  executionSettings,
  buildOptions = NULL,
  saveName = NULL,
  savePath = here::here()
)
```

Arguments

tableShell	The TableShell object to used for generation
executionSettings	The ExecutionSettings object used to generate table shell
buildOptions	The BuildOptions object used to generate table shell
saveName	The name of the table shell sql file
savePath	the folder location to save the file

Value

A sql file written to a specific location

saveTableShellResults	<i>Function that previews sql script used to generate results for table shell</i>
-----------------------	---

Description

Function that previews sql script used to generate results for table shell

Usage

```
saveTableShellResults(result, saveName, savePath = here::here())
```

Arguments

result	the list output from generateTableShell containing a categorical and continuous tibble
saveName	The save name of the csv files
savePath	the folder location to save the csv files

Value

A sql file written to a specific location

Score	<i>Score Statistic</i>
-------	------------------------

Description

A statistic that converts a categorical value to a continuous value by modifying the occurrence of an event by a weight and summing across patients.

Super class

`ClinicalCharacteristics::Statistic` -> Score

Active bindings

`weight` a numeric value to modify the value of an occurrence

Methods**Public methods:**

- `Score$new()`
- `Score$clone()`

Method new():

Usage:

`Score$new(personLine, weight)`

Arguments:

`personLine` the means of converting occurrences to a single event per patient. For a score currently only enabled for any occurrence

`weight` a numeric value to modify the value of an occurrence

Method clone(): The objects of this class are cloneable with this method.

Usage:

`Score$clone(deep = FALSE)`

Arguments:

`deep` Whether to make a deep clone.

sourceConceptSet	<i>Function to create a source concept set</i>
------------------	--

Description

Function to create a source concept set

Usage

```
sourceConceptSet(sourceConceptTable, name)
```

Arguments

sourceConceptTable	a dataframe with source concepts from the OMOP vocabulary
name	the name of source concept set

Value

a SourceConceptSet R6 class specifying the source concepts in use

SourceConceptSetLineItem	<i>SourceConceptSetLineItem</i>
--------------------------	---------------------------------

Description

An R6 class to define a SourceConceptSetLineItem

Super class

```
ClinicalCharacteristics::LineItem -> SourceConceptSetLineItem
```

Methods**Public methods:**

- `SourceConceptSetLineItem$new()`
- `SourceConceptSetLineItem$grabSourceConceptSet()`
- `SourceConceptSetLineItem$clone()`

Method new():

Usage:

```
SourceConceptSetLineItem$new(
  sectionLabel,
  domainTable,
  sourceConceptSet,
  timeInterval,
  statistic
)
```

Arguments:

`sectionLabel` a label for the table shell section

`domainTable` the domain table in the cdm

`sourceConceptSet` a source concept Set

`timeInterval` a time interval class object to determine the time frame to consider the analytic

`statistic` a Statistic Class object used to determine what type of analytic should be done for the line item

Method `grabSourceConceptSet()`: retrieve the source concept set

Usage:

`SourceConceptSetLineItem$grabSourceConceptSet()`

Method `clone()`: The objects of this class are cloneable with this method.

Usage:

`SourceConceptSetLineItem$clone(deep = FALSE)`

Arguments:

`deep` Whether to make a deep clone.

Statistic

An R6 class to define a Statistic object

Description

A Statistic is a type of metric to be used for characterization. Specific types of statistics are defined in derived classes

Methods**Public methods:**

- `Statistic$new()`
- `Statistic$getStatisticType()`
- `Statistic$getAggregationType()`
- `Statistic$getPersonLineTransformation()`
- `Statistic$getBreaksIfAny()`
- `Statistic$getWeightsIfAny()`
- `Statistic$clone()`

Method `new()`:*Usage:*

`Statistic$new(statType, personLine, aggType)`

Arguments:

`statType` the type of statistic

`personLine` the means of converting occurrences to a single event per patient

`aggType` the way the metric is reported either categorical or continuous

Method `getStatisticType()`: retrieve the statistic type

Usage:

Statistic\$getStatisticType()

Method getAggregationType(): retrieve the aggregation type

Usage:

Statistic\$getAggregationType()

Method getPersonLineTransformation(): retrieve the person line transformation

Usage:

Statistic\$getPersonLineTransformation()

Method getBreaksIfAny(): retrieve the breaks object from the statistic object

Usage:

Statistic\$getBreaksIfAny()

Method getWeightsIfAny(): retrieve the weights object from the statistic object

Usage:

Statistic\$getWeightsIfAny()

Method clone(): The objects of this class are cloneable with this method.

Usage:

Statistic\$clone(deep = FALSE)

Arguments:

deep Whether to make a deep clone.

TableShell

Table Shell

Description

An R6 class to define a TableShell object

Methods

Public methods:

- [TableShell\\$new\(\)](#)
- [TableShell\\$getTitle\(\)](#)
- [TableShell\\$getTableShellMeta\(\)](#)
- [TableShell\\$getTargetCohorts\(\)](#)
- [TableShell\\$getLineItems\(\)](#)
- [TableShell\\$printJobDetails\(\)](#)
- [TableShell\\$buildTableShellSql\(\)](#)
- [TableShell\\$outputResults\(\)](#)
- [TableShell\\$dropTempTables\(\)](#)
- [TableShell\\$clone\(\)](#)

Method new():

Usage:

TableShell\$new(title, targetCohorts, lineItems)

Arguments:

title the title of the table shell

targetCohorts a list of CohortInfo class objects that describe the index cohorts

lineItems a list of line item class objects

Method getTitle(): get the title of the table shell

Usage:

TableShell\$getTitle()

Method getTableShellMeta(): get the meta information for the table shell build

Usage:

TableShell\$getTableShellMeta()

Method getTargetCohorts(): get the target cohorts from the table shell

Usage:

TableShell\$getTargetCohorts()

Method getLineItems(): get the lineItems from the table shell

Usage:

TableShell\$getLineItems()

Method printJobDetails(): print the job details of the table shell

Usage:

TableShell\$printJobDetails()

Method buildTableShellSql(): function creates the table shell sql needed for the execution

Usage:

TableShell\$buildTableShellSql(executionSettings, buildOptions)

Arguments:

executionSettings an executionSettings class obj

buildOptions a buildOptions class obj

Method outputResults(): retrieves results from dbms and formats for review

Usage:

TableShell\$outputResults(executionSettings, buildOptions)

Arguments:

executionSettings an executionSettings class obj

buildOptions a buildOptions class obj

Method dropTempTables(): drop all temp tables from the tableShell build

Usage:

TableShell\$dropTempTables(executionSettings, buildOptions)

Arguments:

executionSettings an executionSettings class obj

buildOptions a buildOptions class obj

Method clone(): The objects of this class are cloneable with this method.

Usage:

```
TableShell$clone(deep = FALSE)
```

Arguments:

deep Whether to make a deep clone.

timeInterval	Create a single time interval
--------------	-------------------------------

Description

Create a single time interval

Usage

```
timeInterval(lb, rb)
```

Arguments

lb	the left bound of the time interval
rb	the right bound of the time interval

Value

A time interval object

TimeIntervalClass	<i>TimeIntervalClass</i>
-------------------	--------------------------

Description

An R6 class to define a TimeIntervalClass

Methods

- Public methods:**
- `TimeIntervalClass$new()`
 - `TimeIntervalClass$getLb()`
 - `TimeIntervalClass$getRb()`
 - `TimeIntervalClass$getTimeLabel()`
 - `TimeIntervalClass$getTimeInterval()`
 - `TimeIntervalClass$clone()`

Method new():

Usage:

```
TimeIntervalClass$new(lb, rb)
```

Arguments:

lb left bound - the start of the time interval
 rb right bound - the end of the time interval

Method getLb(): return the left bound

Usage:

TimeIntervalClass\$getLb()

Method getRb(): return the right bound

Usage:

TimeIntervalClass\$getRb()

Method getTimeLabel(): create and return time labels for left and right bounds

Usage:

TimeIntervalClass\$getTimeLabel()

Method getTimeInterval(): return a tibble with the left and right bounds

Usage:

TimeIntervalClass\$getTimeInterval()

Method clone(): The objects of this class are cloneable with this method.

Usage:

TimeIntervalClass\$clone(deep = FALSE)

Arguments:

deep Whether to make a deep clone.

timeToFirst

Time To First

Description

Create a time to stat where any occurrence is valid

Usage

```
timeToFirst()
```

Value

A stat object continuousDistribution

`yearlyRate`*Create a yearly interval rate statistic*

Description

This statistic sums the number of occurrences of an event in a `timeInterval` and divides it by the time (modified by year) to construct a rate per patient. This can then be summarized as a continuous variable

Usage

```
yearlyRate()
```

Value

A stat object of class `intervalRate`

Index

[addDefaultEthnicityLineItems](#), [3](#)
[addDefaultGenderLineItems](#), [3](#)
[adherentPresenceStat](#), [4](#)
[age10yrGrp](#), [4](#)
[age5yrGrp](#), [4](#)
[ageCharBreaks](#), [5](#)
[ageCharCts](#), [5](#)
[anyCountBreaksStat](#), [5](#)
[anyCountCtsStat](#), [6](#)
[anyPresenceStat](#), [6](#)
[anyScore](#), [6](#)

[Breaks](#), [7](#)
[BreaksStrategy](#), [7](#)
[BuildOptions](#), [8](#)

[ClinicalCharacteristics::LineItem](#),
 [11–13](#), [27](#), [45](#)
[ClinicalCharacteristics::Statistic](#), [7](#),
 [14](#), [24–26](#), [28–30](#), [34](#), [41](#), [44](#)
[cohortFollowupTime](#), [10](#)
[CohortInfo](#), [10](#)
[CohortLineItem](#), [11](#)
[ConceptSetGroupLineItem](#), [12](#)
[ConceptSetLineItem](#), [13](#)
[ContinuousDistribution](#), [14](#)
[createCohortInfo](#), [15](#)
[createCohortLineItem](#), [15](#)
[createCohortLineItemBatch](#), [16](#)
[createConceptSetGroupLineItem](#), [16](#)
[createConceptSetLineItem](#), [17](#)
[createConceptSetLineItemBatch](#), [18](#)
[createDemographicLineItem](#), [19](#)
[createExecutionSettings](#), [19](#)
[createSourceConceptSetLineItem](#), [20](#)
[createSourceConceptSetLineItemBatch](#),
 [21](#)
[createTableShell](#), [21](#)

[defaultTableShellBuildOptions](#), [22](#)
[defaultYearGrp](#), [23](#)
[DemographicAge](#), [24](#)
[DemographicCohortTime](#), [25](#)
[DemographicConcept](#), [25](#)

[DemographicIndexYear](#), [26](#)
[DemographicLineItem](#), [27](#)
[DemographicLocation](#), [28](#)
[DemographicPayerType](#), [29](#)
[DemographicRace](#), [30](#)

[ExecutionSettings](#), [31](#)

[femaleGender](#), [32](#)

[generateTableShell](#), [33](#)

[indexYear](#), [33](#)
[IntervalRate](#), [34](#)

[LineItem](#), [34](#)
[lineItems](#), [36](#)
[lookupSourceConcepts](#), [36](#)

[maleGender](#), [37](#)
[monthlyRate](#), [37](#)

[newConceptBreaks](#), [37](#)
[newValueBreaks](#), [38](#)

[observedCountBreaksStat](#), [38](#)
[observedCountCtsStat](#), [39](#)
[observedPresenceStat](#), [39](#)
[observedTimeToFirst](#), [39](#)

[parseCohortInfoFromDf](#), [40](#)
[payerType](#), [40](#)
[personLocation](#), [41](#)
[Presence](#), [41](#)

[quanCharlsonComorbidityScore](#), [42](#)

[raceCategory](#), [42](#)
[reviewTableShellSql](#), [43](#)

[saveTableShellResults](#), [43](#)
[Score](#), [44](#)
[sourceConceptSet](#), [45](#)
[SourceConceptSetLineItem](#), [45](#)
[Statistic](#), [46](#)

TableShell, [47](#)
timeInterval, [49](#)
TimeIntervalClass, [49](#)
timeToFirst, [50](#)

yearlyRate, [51](#)